

Application of GIS in the Planning, Response and Recovery Stages of the Disaster Response and Management of Wildfires

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Introduction

Disasters are characterized by the scope of an emergency, with an emergency situation becoming a disaster when it exceeds the ability of local resources to manage it [9]. Disasters can be a variety of things that resulted from a combination of factors, from man-made to natural hazards or both. However, with climate change, urban growth, and lack of preparedness for these disasters, it is causing these events to become worse, increasing the impacts on life and economic loss [1].

Wildfires are an example of a disaster that can be caused from both natural phenomenon and human actions, or a combination of both. They are unplanned and uncontrolled fire that burns in a variety of environments such as in forests, grasslands, or prairies. It can happen anytime and anywhere, not limited to specific locations [7, 5].

There are three types of wildfires – ground fires, surface fires and crown fires. Ground fires can ignite in soils rich with organic matter which will fuel the flames. These can smolder for a long time and under the correct conditions, can grow. Surface fires burn in dry vegetation that is lying on the ground or directly above while crown fires burn through the leaves of shrubs and tree canopies. Areas can experience one or all three types of wildfires [5].



Figure 1: Image of the three of the wildfire types that can be seen in nature - Ground, Surface and Crown [12].

Due to climate change increasing temperatures and dryer conditions in some areas, more ecosystems have become more susceptible to wildfires, with the frequency and size of wildfires increasing [7, 5]. In southwestern United States, the average wildfire season is now three months longer than it was a few decades ago [6]. There are a variety of situations that can lead to wildfires (Figure 2), from lightening to human interference, and affect how it will burn. Leaving campfires unattended or fireworks can cause wildfires to begin, and combined with natural effects, such as wind direction, temperature, dry forests can cause them to get out of hand. From California to Canada to the Siberian arctic, there are now a record-breaking number of wildfires per year [6].

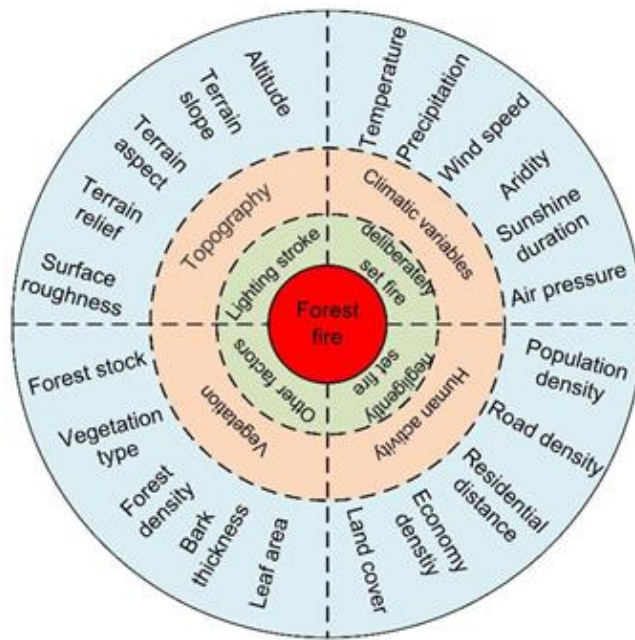


Figure 2: Visualization of the situations and variables that can impact wildfires. Green ring discusses and event to cause a fire, the red ring describes what variables can impact the fire, while the blue ring is more detailed attributes that can be used to analyze a wildfire [8].

Wildfires can lead to a number of issues, both to the environment and society. If they grow out of control they can disrupt transportation routes, communication networks, power and gas services as well as water supplies [7]. They can destroy crops and resources, kill animals and people, impacting the weather, climate and health of people and animals due to the large quantities of carbon dioxide, carbon monoxide and particulate matter released into the atmosphere.

GIS Applied to the Disaster Response/Management of Wildfires

To help manage wildfires as well as other disasters, disaster response and emergency management practices have been put into place to try and minimize the impacts of these events on both the communities as well as the environment. These practices and programs gather data from a wide variety of sources, organize it and provide visual displays to understand the scope of the disaster or emergency [6], applying it to the five main phases of a disaster response plan. The five phases are used to protect life, property, and the environment – planning, preparation, mitigation, response, and recovery [9].

With wildfires, applying GIS to the disaster response plan can allow for the tracking of fires through aerial and satellite imagery, an easy way to provide the public the locations of the various fires and their statuses through maps as well as combine weather, temperature, and vegetation data to try and predict where fires will start. Regardless of the part of the wildfire cycle that is being dealt with, GIS can provide tools to identify, analyze the landscape so that when a fire breaks out, maps are available help decision makers understand the situation and what is needed to reduce the negative consequences of a wildfire [1]. GIS techniques applied to the Planning, Response and Recovery Phases are vital in managing a wildfire outbreak – determining the best methods for predicting, monitoring, and fighting and recovery after a fire.

Planning

Within the planning phase, the hazards, risks, area preparedness and response plans are assessed. Everything is done to analyze what type of disaster or emergency will happen, when it will happen as well as the positive or negative impacts. GIS can aid in compiling data into one location, allowing various groups to view and use, as well as create maps which can help to locate the hazard, what risks and consequences it poses to the surrounding area [9].

Applied to wildfires, with GIS, we can look at past data to create a map showing past wildfire data in the area to try and predict the size, location, intensity, and movement of a potential new fire. Due to the unpredictable nature, past data can aid in narrowing down locations where fires may break out and areas where they will not. Remote sensing data can be used to look at hot spots and temperature data in the study area, both for the current month and previous years to try and determine any patterns for the areas as the probability of wildfires increases as temperatures increase [8]. This can help to predict the number of fires and their potential severity since with higher temperatures there can be environmental effects, such as drought, that can affect the vegetation that can fuel a wildfire.

Maps can also be created to display land use and vegetation types for an area, determining which areas will be potential areas a wildfire can start, and which are most affected if a wildfire moves in that direction, ex. If a wildfire grows towards a town, as well as which direction it will most likely travel based on the amount of vegetation there is to act as fuel for the fire. Vegetation analysis provides key information on what will ultimately fuel the wildfire. In the US, the National Fire Danger Rating System (NFDRS) has created five fuel classes based on which vegetation combination is the most flammable [2]. The Ministry of Natural Resources Canada has also broken down fuel types into classes to better understand the vegetation. This can be used to create fuel models to predict fire behaviour which can help with mitigation as well as indicate where preventative controlled burns, thinning of forests and removal of loose leaves and plant debris are beneficial to reduce potential wildfire fuel [3].

Figure 3 below is an example of a fire prediction map based off the vegetation type in the area for Pelican Mountain Research Site, near Slave Lake Alberta. A represents the predicted fire behaviour and what fuels there are in that section. In this case it is Boreal Spruce (black spruce trees), classified as C-2 Boreal Spruce Fuel Type. Section B shows a more detailed look into the vegetation cover, what is actually black spruce versus what is not, as well as the density of the trees and vegetation. The three pictures on the right hand side are a real life example of the canopy densities in the map in part B. This can help to understand where the fire may become more intense and burn faster (in the higher density areas) and start to visualize a plan to minimize the spread of a fire in this location [13].

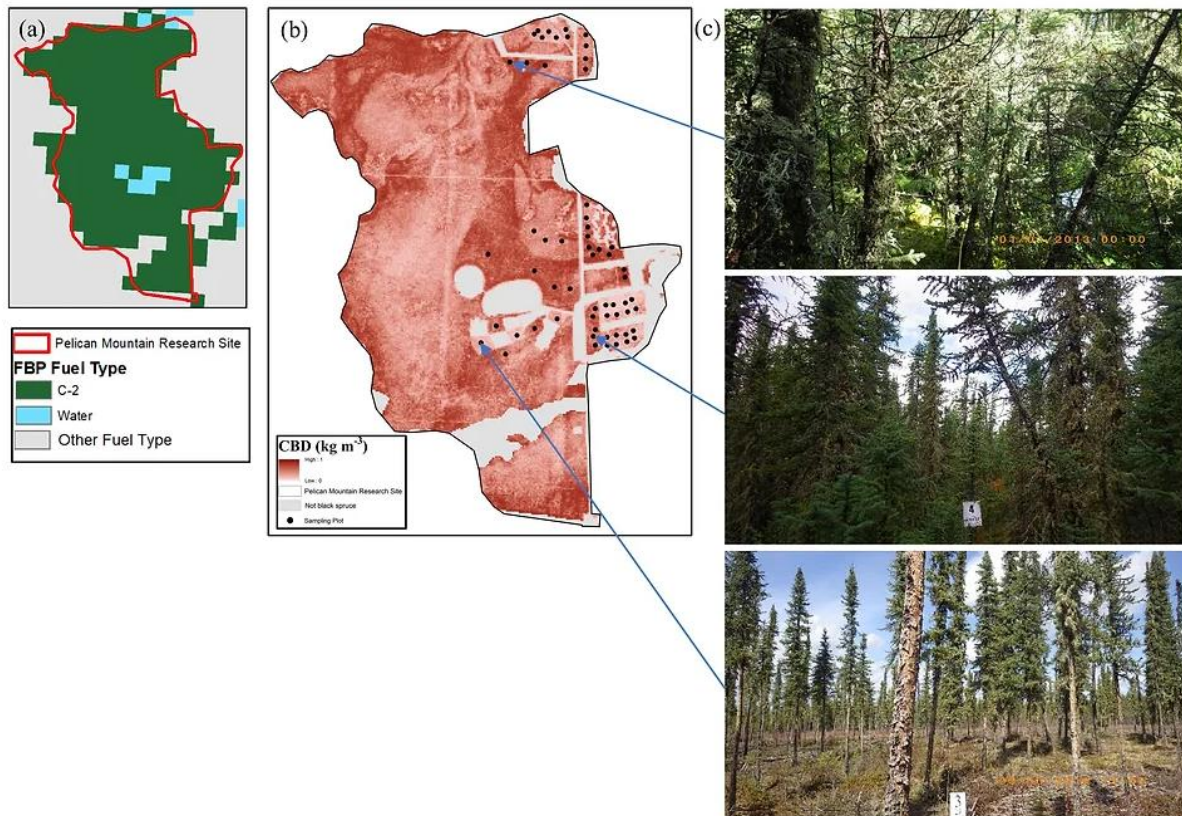


Figure 3: Displays the use of vegetation type and canopy density to analyze the fire behaviour for the study area of Pelican Mountain Research Site in Alberta, Canada [13].

Vegetation and land use maps can also be used to look at the consequences to the environment the fire will have – which vegetation types will be gone, which water supplies will become contaminated, which animal will lose their habitats and become displaced, which towns could become cut off from outside help and resources. This can also be combined with climate data (ex. Precipitation, wind speeds), soil data as well as topography (ex. Slope data). Precipitation data can be analyzed and then predicted to determine if rain will be a possibility if a fire breaks out to help combat or if the ground will be wet in areas, slowing down the growth of the fire. Determining the topography of the area can help in determining which way the fire may flow, for example, if a town in the mountains is uphill from a potential break out location for a forest fire, evacuation plans and routes may want to be created since wildfire move quicker uphill than they do down hill [5].

Maps can also be created using GIS to plan where the location of fire crews will be stationed through out the wildfire seasons. Data layers can be added to the maps to show the past extents of the fires before they were extinguished or gotten under control as well as the response times of the previous year's crews to the area. Combine this data together, fire fighter crew locations can be moved around so that areas can be covered better, and/or more people moved to certain areas that will require more help if a wildfire breaks out.

Using GIS, evacuation routes can be planned and provided to the residents of the areas affected. The number of households in an area can major roadways can be factored together to determine which communities use which roads to try and minimize traffic. Populations of town in the wildfire affected areas can be analyzed to determine if any will become

overwhelmed or need additional resources if evacuees end of sheltering there, allowing for the planning of having addition food, water, and shelter resources on standby.

Response

The response phase concerns all the activates and actions that take place right after the emergency or disaster begins. The situation is analyzed, attempted to be brought under control, aide is provided to those affected and the probability of a secondary damage is minimized. GIS can be an important part of this stage since it can determine the extent of the issue as well as monitor any changes [9].

With wildfires, interactive maps can be created that show the extent of the wildfire, its current status and any other important information that will be needed for that area, ex. Restrictions in place or evacuation areas. The government of Canada has a section on their website that has an interactive map for fires across Canada as well as a collection of maps showing the current and previous year's hotspots (which can be used to look at where fires are). The governments of BC and Ontario also have interactive maps showing the locations, extents and status of the wildfires, evacuation status and restricted areas (Figures 4 and 5, 6) These can be used by the public and wildfire fighters to know what is going on in the area, if they will be impacted and which areas to stay away from, minimizing the risk of people coming across an out-of-control fire which can cause harm to their life. These are updated daily based on information provided and gathered from technicians on the ground. On the BC interactive map, they provide statistics on the fires for the year (Figure 4) while the Ontario interactive maps has an option to add the fire stations/management headquarters (Figure 6).

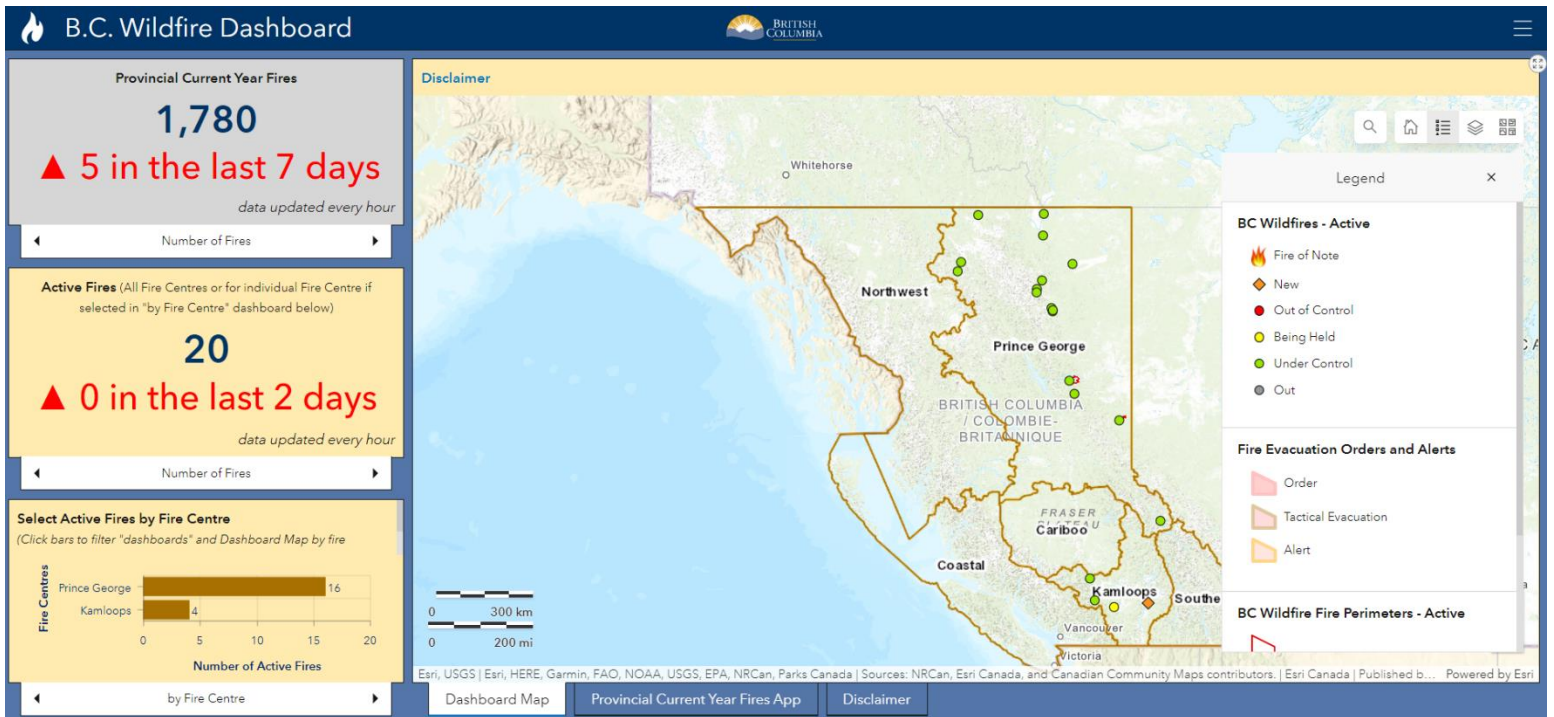


Figure 4: Still photo of the interactive map found on the Government of BC website to display the wildfires and their status for the BC area. It also provides information on evacuation notices and active wildfire areas [10].

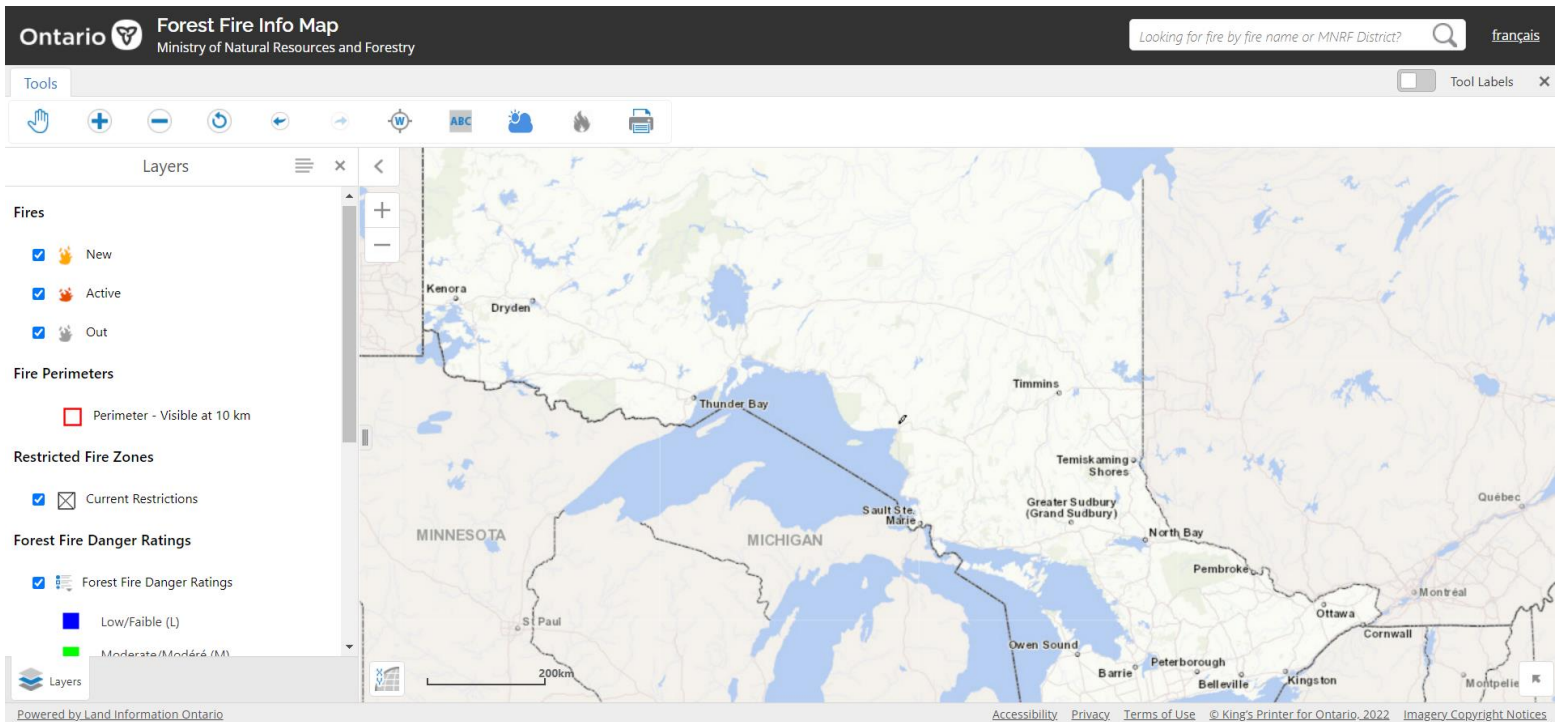


Figure 5: Still photo of the interactive map on the Government of Ontario website. Will display the location and status of wildfires as well as the restricted zones. It is a current image of the map as of March 23, 2023, so no wildfires are currently active [11].

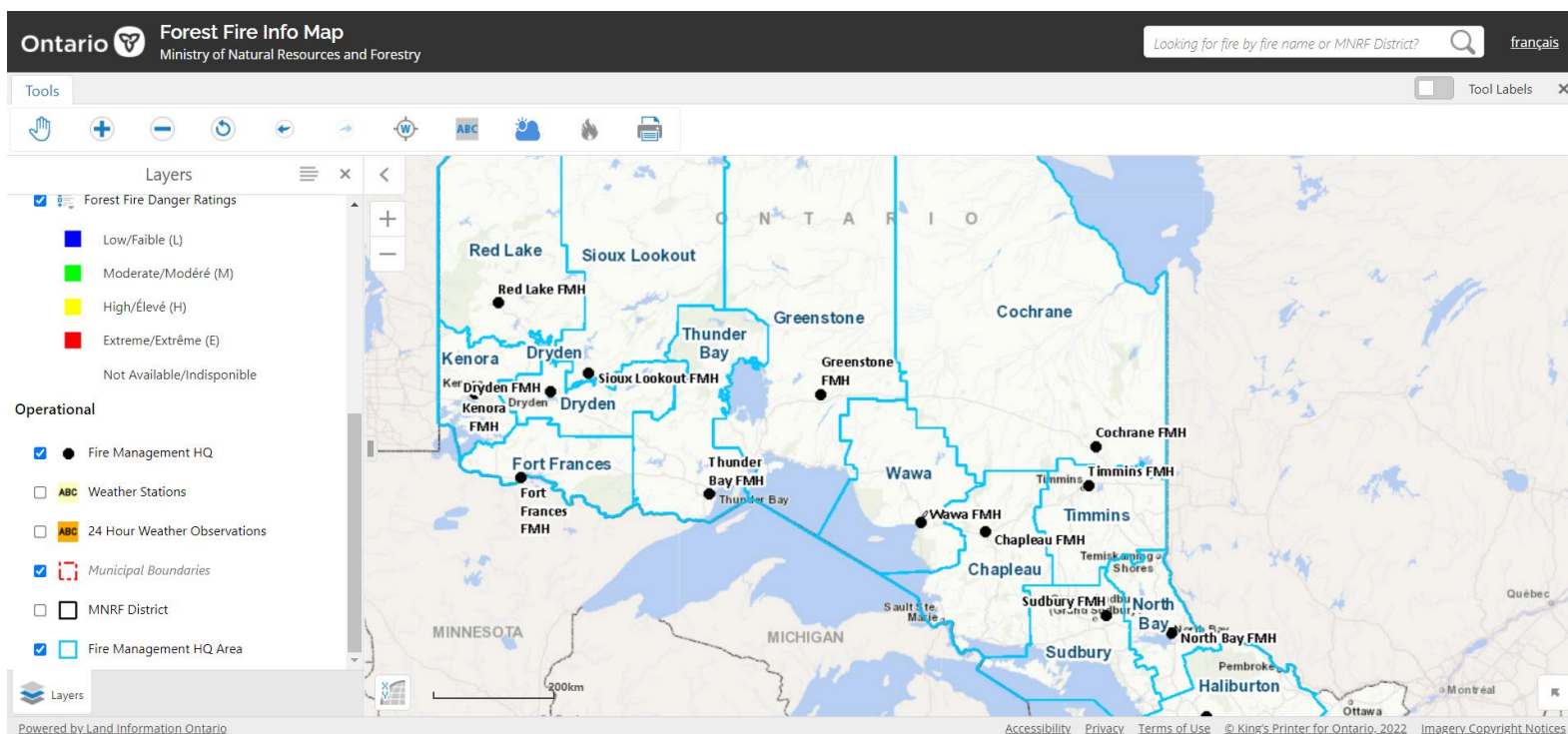


Figure 6: Still photo of the interactive map on the Government of Ontario website. Will display the location and status of wildfires as well as the restricted zones. It is a current image of the map as of March 23, 2023, so no wildfires are currently active. The fire management HQ locations and areas are highlighted on the map [11].

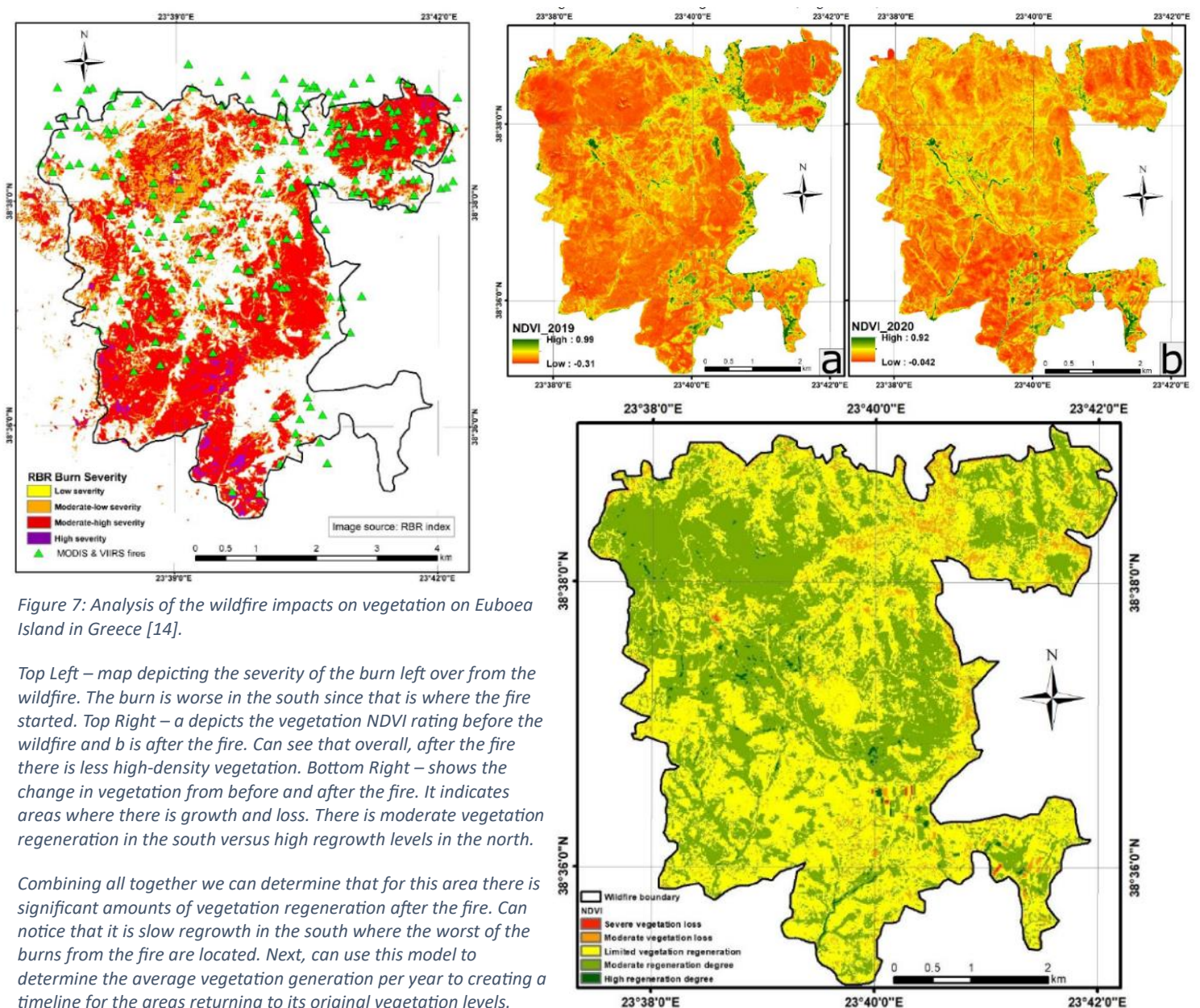
Aerial photographs can also be used to look at the damage caused by the wildfires to vegetation, habitats, infrastructure, and roadways. In boreal forests of Alaska, satellite imagery was used to determine the area of fires by looking at the levels of the Normalized Difference vegetation index (NDVI) from the Advanced Very High-Resolution Radiometer (AVHRR), which are lower where fires have damaged the plant canopy [4]. They can be used to start to compile information on the severity of the damage and how long it may take for the area to recover, including the amount of effort required. The remotely sensed data can also provide a look at the temperatures of the area. Using the AIRDAS thermal line scanner, the four bands can provide information on the temperatures of the fire and the areas around it. Band 1, the visible band, can monitor smoke plumes and distinguish between surface features that are not obscured by smoke or clouds [2]. Band 2 can be used to analyze vegetation composition and the fires at temperatures above 300 degrees Celsius, while being able to penetrate through most smoke plumes [2]. Band 3 is specifically for capturing information on the temperatures of the fires through their smoke while band 4 collects the thermal data on earth, focusing on the soil heating behind fire front [2]. This can help to tell you the more dangerous parts of fires, help categorize severity based off temperatures, indicate areas where the fire may not be fully extinguished with a possibility for a re ignition as well as areas that are now cool enough for recovery crews to go in an assess.

Recovery

The final phase of disaster response and emergency management is the recovery. This is when immediate threat to life, property and the environments is not as severe [9]. This is where plans and procedures to return the affected systems to normal, or even better [9].

Recovery can be split into two different categories, short term, and long term. Short term recovery focus on quickly implementing the return of various systems to their minimum operation standard whereas long term recovery can continue for months to years, as life returns to the operating standard of before the disaster or event better [9]. GIS is important in the recovery phase as it can be used to track the changes after the disaster. Applied to wildfires, GIS can be used to track the environmental changes over time, both short term and long terms.

For short term recovery, GIS and remotely sensed data can be used to locate and determine the extent of the damage from the fire. How much vegetation is left over (Figure 7), which industrial systems have been damaged (ex. electrical wires), which areas will need the most assistance. Maps can be made to show which areas the damage has already been cleaned up and services essential to human life have restored so that the recovery personnel can be reallocated to higher priority locations [9].



GIS can also be used to track the amount of particulate matter in the air as well as the increase in greenhouse and trace gases which can alter regional microclimates, nutrient levels, and air quality [8]. This can be modelled to show the levels across the areas surrounding the fires to know which areas will have the worst air quality and how many people need to be aware of these issues. Wind data can be combined with this data to determine which way the particulate matter will blow and how it will affect the people in the areas, the vegetation, and animals. Over time, this can be monitored to see if the levels are dispersing or if it has had a long-term effect on the previous wildfire zone.

With long term recovery, GIS can play a huge role, especially when looking at the vegetation responses after a wildfire. The land cover data can be obtained and then analyzed to look at the percent of vegetation left in an area. The amount of damage can be calculated and compared to previous years data on vegetation regrowth in that area and for the species (Figure 7). This can be used to determine a timeline for the natural environment to return to normal as well as a change detection study to be carried out. This data can also be used to know where human inference with helping plant trees and re grow forests will be needed.

Conclusion

GIS is an important tool to use for disaster response and emergency management. It can have a variety of uses for various events. With wildfires, it provides helpful information on the extent of fires, the damage created as well as an interactive monitoring map to provide information regarding the fires to the public. GIS will not be able to stop all wildfires from happening but it can be used to help prevent serious damage from occurring by aiding in the prediction of wildfire breakout location based on temperature, vegetation and topographical data, help to minimize unnecessary interactions of the general public and wildfires with frequently updated maps showing the location of the fires in the areas as well as aide in the recovery after a fire.

With disaster and emergency responses it is important in these situations to have the right data at the right time [1] and GIS can be useful for analyzing data, displaying information in a user-friendly manner, ex. create maps, as well as share information between departments, bringing together critical information [1].

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